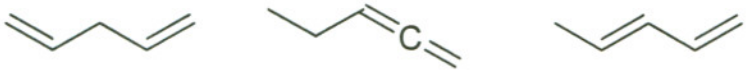
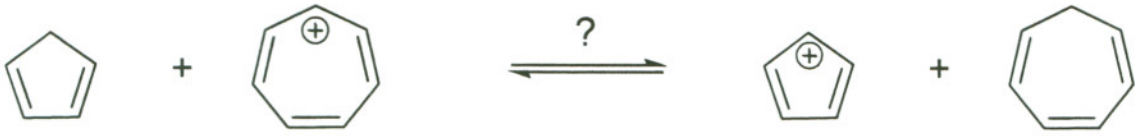
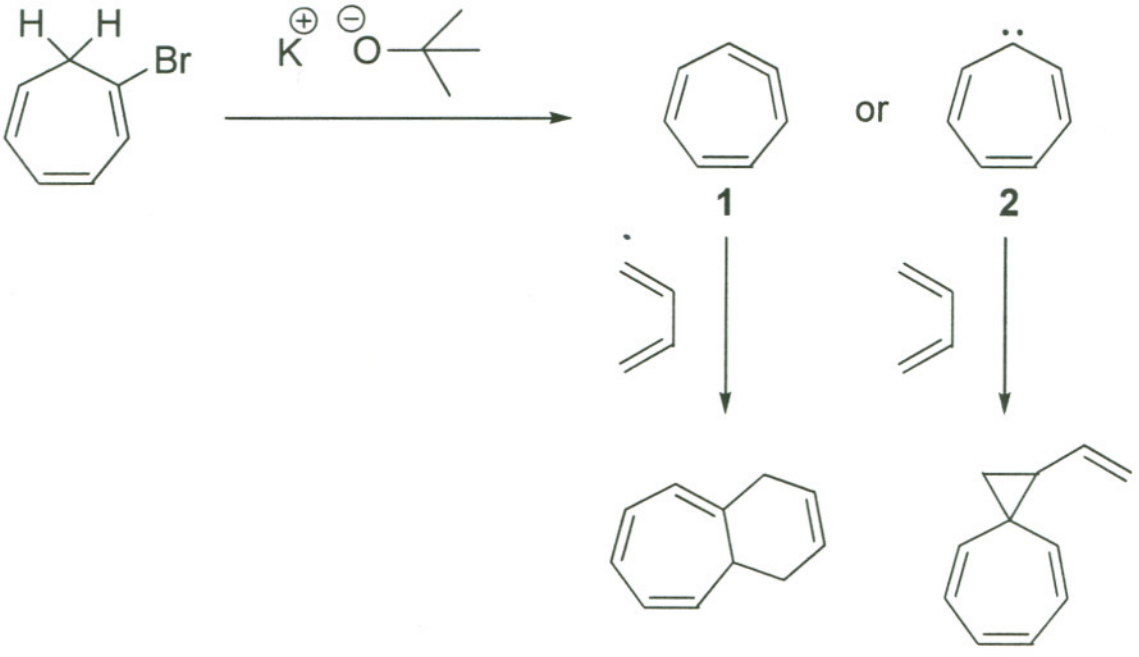
Organic cumulative exam, November 20, 2009

1. (20 points) Rank the following dienes in terms of stability (most stable to least stable). Justify your answer.



2. (20 points) Indicate whether the following equilibrium lies toward the reactants or products and justify your answer.



3. (60 points) When 1-bromocycloheptatriene is introduced to a strong base, it was thought that either cycloheptatetraene (**1**) or cycloheptatrienylidene (**2**) is initially generated. Each of these intermediates can be trapped. A typical example of that is as shown below.
Show how by use of a chiral base, one could make a convincing argument that would distinguish between one of the two intermediates being generated.
If cycloheptatetrene **1** is generated initially with the chiral base but then an equilibrium is established between the **1** and **2**, show how this would affect your argument in part a.